Order-Convex Subsets of Grid Posets: A New Exponential Combinatorial Class

with Structural Classification, Transfer-Matrix Enumeration, and a Supermultiplicativity Theorem

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Abstract

We study order-convex subsets of the product poset $[m] \times [n]$, where $[m] = \{0, 1, \ldots, m-1\}$ is the chain of m elements and the product carries the component-wise partial order. We prove the *Endpoint Classification Theorem*: a subset S is order-convex if and only if each row fiber is an interval and the endpoints satisfy a rectangle-fill condition. Convex subsets decompose uniquely into maximal contiguous row blocks, with blocks separated by gaps across which no comparable pairs exist (strict anti-monotonicity). Using this decomposition, we establish a *Supermultiplicativity Theorem*: the counting sequence $a_m := |\mathrm{CC}([m]^2)|$ satisfies $a_{m+n} \geq a_m \cdot a_n$ for all $m, n \geq 1$. By Fekete's lemma, this implies the growth constant $\rho := \lim_{m\to\infty} a_m^{1/m}$ *exists* and is finite ($\rho \leq 16$, via an ideal/filter injection giving $a_m \leq \binom{2m}{m}^2$). A transfer-matrix dynamic program with $O(n^2)$ boundary parameters enables computation of 50 exact terms of the sequence $a_m = 2, 13, 114, 1146, 12578, 146581, \ldots$, confirming the growth constant $\rho = 16$ (the upper bound $\rho \leq 16$ is proved via the ideal/filter injection into pairs of antitone functions counted by $\binom{2m}{m}$; the matching lower bound uses a Lindström-type crossing-pairs bound yielding $|\mathrm{CC}([m]^2)| \geq \binom{2m}{m}^2/(2(m+1))$, squeezed to equality by Fekete's lemma). This exceeds the nearest generalized Fuss–Catalan benchmark $6^6/5^5 \approx 14.93$ by a factor of approximately 1.07, and the ratio $a_m/\mathrm{FC}(m)$ diverges exponentially. We conclude that the growth constant is $\rho = 16 = 2^4$ exactly; since $16 > 6^6/5^5 \approx 14.93$, this sequence does not belong to the $(6, 2)$ generalized Fuss–Catalan family, despite matching its first three terms. All structural theorems, the supermultiplicativity, and the growth constant existence are formally verified [11].

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1 Introduction

A subset S of a partially ordered set P is *order-convex* (or simply *convex*) if whenever $a, b \in S$ with $a \leq b$, every element c with $a \leq c \leq b$ also belongs to S. Order-convex subsets arise naturally in causal set theory [3], lattice theory [1], and combinatorial optimization [5]. The number $|\mathrm{CC}(P)|$ of convex subsets is a fundamental invariant of the poset P, and the *Noetherian ratio* $\gamma(P) := |\mathrm{CC}(P)|/|\mathrm{Int}(P)|$ measures geometric regularity [4].

The simplest infinite family of posets exhibiting non-trivial convexity structure is the *grid poset* $[m] \times [n]$, the product of two chains with the component-wise partial order:

$$(i_1, j_1) \leq (i_2, j_2) \quad \Longleftrightarrow \quad i_1 \leq i_2 \text{ and } j_1 \leq j_2.$$

This paper studies the counting sequence $a_m := |\mathrm{CC}([m]^2)|$—the number of order-convex subsets of the $m \times m$ grid.

1.1 Motivation

The counting problem arises from the causal-algebraic geometry framework of [4], where the *Noetherian ratio* $\gamma(P) = |\mathrm{CC}(P)|/|\mathrm{Int}(P)|$ serves as an invariant that is strictly finer than all classical poset invariants and detects manifold-likeness in causal set theory. A *width scaling law* $\gamma = O(n^{2(w-1)})$ was proved for arbitrary posets, but the generic bound is orders of magnitude loose on structured families such as grids. Understanding the exact combinatorics of $|\mathrm{CC}([m]^2)|$ is the first step toward sharp asymptotics for the Noetherian ratio on d-dimensional discrete spacetimes.

1.2 Main results

Theorem 1.1 (Endpoint Classification). *A subset $S \subseteq [m] \times [n]$ is order-convex if and only if:*

1. *Each row fiber $S \cap (\{i\} \times [n])$ is an interval $[L_i, R_i]$ (or empty).*
2. *For all active rows $i_1 < i_2$ with $L_{i_1} \leq R_{i_2}$, every intermediate row k with $i_1 \leq k \leq i_2$ is active with $L_k \leq L_{i_1}$ and $R_k \geq R_{i_2}$.*

Theorem 1.2 (Block Decomposition). *Every order-convex subset of $[m] \times [n]$ decomposes uniquely into maximal contiguous active-row blocks. Distinct blocks are strictly anti-monotone: if block B_1 occupies rows above block B_2, then every element of B_1 is incomparable with every element of B_2 in the product order.*

Theorem 1.3 (Supermultiplicativity). *For all $m, n \geq 1$,*

$$|\mathrm{CC}([m+n]^2)| \;\geq\; |\mathrm{CC}([m]^2)| \cdot |\mathrm{CC}([n]^2)|.$$

Corollary 1.4 (Existence of Growth Constant). *The limit*

$$\rho \;:=\; \lim_{m\to\infty} |\mathrm{CC}([m]^2)|^{1/m} \;=\; \sup_{m\geq 1} |\mathrm{CC}([m]^2)|^{1/m}$$

exists and equals $\rho = 16 = 2^4$.

Proof. Supermultiplicativity (Theorem 1.3) gives superadditivity of $\log a_m$. For finiteness: every order-convex $S \subseteq [m]^2$ is determined by the pair $(I(S), F(S))$ where $I(S)$ is the downward closure and $F(S)$ is the upward closure, since $S = I(S) \cap F(S)$ by convexity. The map $S \mapsto (I(S), F(S))$ is injective. The number of order ideals of $[m]^2$ is $\binom{2m}{m}$ (monotone lattice paths from $(0,0)$ to (m,m)). Hence $a_m \leq \binom{2m}{m}^2 \leq 16^m$, giving $\rho \leq 16$. By Fekete's lemma, $\rho = \lim a_m^{1/m} = \sup a_m^{1/m}$. The exact value $\rho = 16$ follows from the matching lower bound $|\mathrm{CC}([m]^2)| \geq \binom{2m}{m}^2/(2(m+1))$ (Theorem 10.2), which gives $\rho \geq 16 \cdot (2(m+1))^{-1/m} \to 16$. □

Proposition 1.5 (Transfer Matrix). *For fixed n, the transfer-matrix dynamic program for $|\mathrm{CC}([m] \times [n])|$ has a state space parameterized by boundary pairs $(L_{\min}, L_{\min,\mathrm{prev}})$, of which $\binom{n+2}{2}$ are reachable (verified computationally for $n \leq 8$). The full state additionally tracks the right-endpoint profile, giving a total state count that grows faster than $\binom{n+2}{2}$ (sequence: 2, 6, 14, 25, 41, 63, 92, 129 for $n = 1, \ldots, 8$).*

Corollary 1.6 (New Exponential Class). *Since $\rho = 16 > 6^6/5^5 \approx 14.93$, the sequence $|\mathrm{CC}([m]^2)|$ does not belong to the $(6,2)$ generalized Fuss–Catalan class. The order-convex subsets of the square grid define a new exponential combinatorial class.*

Remark 1.7. Prior to the proof that $\rho = 16$, numerical evidence from 50 exact terms supported the separation $\rho > 6^6/5^5$ but fell short of a rigorous lower bound: $|\mathrm{CC}([42]^2)|^{1/42} \approx 14.47 < 14.93$. The resolution came from the algebraic direction — proving the ideal/filter injection is asymptotically tight — rather than from improved numerical bounds.

1.3 Relation to prior work

The Noetherian ratio γ and its basic properties were introduced in [4], where the width scaling law $2^w \leq |\mathrm{CC}(C)| \leq (\lceil n/w \rceil^2 + 1)^w$ was proved using Dilworth's theorem and convex factorization. That paper established the general framework; the present paper gives the first exact analysis of a flagship family.

Order-convex subsets of lattices have been studied by Semenova and Wehrung [10] from the lattice-theoretic perspective, and sublattices of products of chains by Grätzer and Quackenbush [7]. Barnette, Nichols, and Richmond [2] gave explicit formulas for the number of convex subsets of $[n] \times [m]$ for small m, establishing the enumeration problem; the present paper extends their work to determine the asymptotic growth constant, prove supermultiplicativity, and establish the transfer-matrix structure for all n.

The generalized Fuss–Catalan numbers $\frac{r}{np+r}\binom{np+r}{n}$ count lattice paths and are tabulated in OEIS [9] as A212071 (for $p = 6, r = 2$) and A386379. The first three terms of our sequence coincide with these, but the sequences diverge at $m = 4$.

2 Definitions and Setup

Definition 2.1. The *grid poset* $[m] \times [n]$ is the set $\{(i, j) : 0 \leq i < m,\ 0 \leq j < n\}$ with the component-wise partial order: $(i_1, j_1) \leq (i_2, j_2)$ iff $i_1 \leq i_2$ and $j_1 \leq j_2$.

Definition 2.2. A subset $S \subseteq [m] \times [n]$ is *order-convex* if for all $a, b \in S$ with $a \leq b$ and all $c \in [m] \times [n]$ with $a \leq c \leq b$, we have $c \in S$. We write $\mathrm{CC}([m] \times [n])$ for the collection of all order-convex subsets, and $|\mathrm{CC}([m] \times [n])|$ for its cardinality.

Definition 2.3. For $S \subseteq [m] \times [n]$ and $i \in [m]$, the *row fiber* is $S_i := \{j \in [n] : (i, j) \in S\}$. Row i is *active* if $S_i \neq \emptyset$.

3 The Endpoint Classification

Lemma 3.1 (Row fibers are intervals)**.** *If $S \subseteq [m] \times [n]$ is order-convex, then each row fiber S_i is a (possibly empty) interval in $[n]$.*

Proof. Suppose $(i, j_1), (i, j_2) \in S$ with $j_1 \leq j_2$. For any j with $j_1 \leq j \leq j_2$, we have $(i, j_1) \leq (i, j) \leq (i, j_2)$ in the product order (same row, columns between). By convexity, $(i, j) \in S$. □

Lemma 3.2 (Rectangle fill)**.** *If S is order-convex and $(i_1, j_1), (i_2, j_2) \in S$ with $i_1 \leq i_2$ and $j_1 \leq j_2$, then $(i, j) \in S$ for all $i_1 \leq i \leq i_2$ and $j_1 \leq j \leq j_2$.*

Proof. Since $(i_1, j_1) \leq (i, j) \leq (i_2, j_2)$ in the product order, convexity gives $(i, j) \in S$. □

Proof of Theorem 1.1. Forward direction. Let S be order-convex. By Lemma 3.1, each row fiber is an interval, say $[L_i, R_i]$ for active rows. Let $i_1 < i_2$ be active rows with $L_{i_1} \leq R_{i_2}$. Then $(i_1, L_{i_1}) \in S$ and $(i_2, R_{i_2}) \in S$ with $(i_1, L_{i_1}) \leq (i_2, R_{i_2})$. By Lemma 3.2, for any k with $i_1 \leq k \leq i_2$ and any j with $L_{i_1} \leq j \leq R_{i_2}$, we have $(k, j) \in S$. In particular, row k is active with $L_k \leq L_{i_1}$ and $R_k \geq R_{i_2}$.

Backward direction. Suppose the endpoint conditions hold. Let $(i_1, j_1), (i_2, j_2) \in S$ with $(i_1, j_1) \leq (i_2, j_2)$, so $i_1 \leq i_2$ and $j_1 \leq j_2$. Since $j_1 \in [L_{i_1}, R_{i_1}]$ and $j_2 \in [L_{i_2}, R_{i_2}]$, we have $L_{i_1} \leq j_1 \leq j_2 \leq R_{i_2}$, so $L_{i_1} \leq R_{i_2}$. By the endpoint condition, every row k with $i_1 \leq k \leq i_2$ is active with $L_k \leq L_{i_1} \leq j_1 \leq j$ and $R_k \geq R_{i_2} \geq j_2 \geq j$ for all $j_1 \leq j \leq j_2$. Hence $(k, j) \in S$ for all $i_1 \leq k \leq i_2$ and $j_1 \leq j \leq j_2$. □

4 The Block Decomposition

Definition 4.1. A *block* of a convex subset S is a maximal contiguous set of active rows. Two blocks are *anti-monotone* if no element of one is comparable to any element of the other.

Proof of Theorem 1.2. Let B_1 and B_2 be distinct blocks with B_1 occupying higher rows (lower indices) than B_2. Since the blocks are separated by at least one empty row, there exists k with B_1 in rows $\leq k-1$ and B_2 in rows $\geq k+1$, with row k empty. Suppose for contradiction that $(i_1, j_1) \in B_1$ and $(i_2, j_2) \in B_2$ are comparable, say $(i_1, j_1) \leq (i_2, j_2)$. Then $i_1 < i_2$ and $j_1 \leq j_2$, so $L_{i_1} \leq j_1 \leq j_2 \leq R_{i_2}$, giving $L_{i_1} \leq R_{i_2}$. By Theorem 1.1, row k must be active—contradicting that row k is empty. □

Proposition 4.2 (Block counting)**.** *The number of contiguous blocks equals the number of comparability-connected components of the active rows.*

Proof. Two active rows in the same contiguous block can always be connected by a chain of comparable pairs (through intermediate rows with overlapping intervals). Two active rows in different blocks cannot be connected by comparable pairs, by Theorem 1.2. □

The block decomposition reveals a *two-sector structure*:

- The **contiguous sector**: subsets with a single block.
- The **gapped sector**: subsets with two or more blocks, separated by anti-monotone gaps.

Table 1 shows the sector fractions for small m.

m	0 blocks	1 block	2 blocks	3+ blocks	Total
1	1	1	0	0	2
2	1	12	0	0	13
3	1	108	5	0	114
4	1	1,030	115	0	1,146
5	1	10,785	1,764	28	12,578
6	1	121,478	24,136	966	146,581
7	1	1,442,392	320,844	20,877	1,784,114
8	1	17,808,426	4,260,321	374,484	22,443,232
9	1	226,558,935	57,015,530	6,147,306	289,721,772
10	1	2,951,472,667	770,955,559	96,361,551	3,818,789,778

Table 1: Sector decomposition of $|\mathrm{CC}([m]^2)|$ by number of contiguous blocks. The empty set (order-convex with no active rows) contributes 1 to the 0-block column.

5 The Supermultiplicativity Theorem

Proof of Theorem 1.3. We construct an injection

$$\mathrm{CC}([m]^2) \times \mathrm{CC}([n]^2) \hookrightarrow \mathrm{CC}([m+n]^2).$$

Given convex $S_1 \subseteq [m]^2$ and $S_2 \subseteq [n]^2$, define embeddings

$$\varphi_1 : [m]^2 \to [m+n]^2, \quad (i,j) \mapsto (i,\ j+n),$$
$$\varphi_2 : [n]^2 \to [m+n]^2, \quad (i,j) \mapsto (i+m,\ j).$$

Set $S := \varphi_1(S_1) \cup \varphi_2(S_2)$.

Both images are convex. The embeddings φ_1, φ_2 are order-preserving and order-reflecting (they preserve and reflect the component-wise partial order). A convex subset mapped by an order-embedding remains convex, since any element between two image points must itself be an image point (by the bounds on row and column indices).

The images are anti-monotone. Every element of $\varphi_1(S_1)$ has column index $\geq n$ and row index $< m$. Every element of $\varphi_2(S_2)$ has column index $< n$ and row index $\geq m$. For $(i_1, j_1 + n) \in \varphi_1(S_1)$ and $(i_2 + m, j_2) \in \varphi_2(S_2)$:

$$j_1 + n \geq n > j_2 \quad (\text{since } j_2 < n),$$

so these elements are incomparable regardless of the row indices.

The union is convex. The union of two convex sets with no comparable pairs between them is convex: if $a \leq c \leq b$ with a, b in the union, then a and b must lie in the *same* image (otherwise they would be comparable across images), and convexity of that image gives c in the same image.

The map is injective. Since $\varphi_1(S_1)$ occupies rows $[0, m)$ with columns $[n, m+n)$ and $\varphi_2(S_2)$ occupies rows $[m, m+n)$ with columns $[0, n)$, the two images are disjoint. Thus S_1 and S_2 are uniquely recoverable from S. □

Proof of Corollary 1.4. Theorem 1.3 gives $a_{m+n} \geq a_m \cdot a_n$, so $\log a_m$ is superadditive. By Fekete's lemma [6], $\lim_{m\to\infty}(\log a_m)/m = \sup_m(\log a_m)/m$, hence $\lim a_m^{1/m} = \sup a_m^{1/m}$. Finiteness: every order-convex $S \subseteq [m]^2$ is determined by the pair $(I(S), F(S))$ of its downward closure and upward closure, since $S = I(S) \cap F(S)$ by convexity. The number of order ideals of $[m]^2$ is $\binom{2m}{m}$ (monotone lattice paths). Hence $a_m \leq \binom{2m}{m}^2 \leq 16^m$, giving $\rho \leq 16$. □

6 Transfer-Matrix Enumeration

The Endpoint Classification (Theorem 1.1) enables a dynamic programming algorithm that constructs convex subsets row by row.

Definition 6.1. A *DP state* for n-column grids is a triple $(\mathbf{c}, L_{\min}, \text{flag})$ where:

- $\mathbf{c} = (c_0, \ldots, c_{n-1}) \in (\{0, \ldots, n-1\} \cup \{\infty\})^n$ is the *constraint tuple*: $c_j = \min\{R_i : L_i \leq j, \text{ row } i \text{ active in current block}\}$, or ∞ if no such row exists;
- $L_{\min} \in \{0, \ldots, n-1\} \cup \{\infty\}$ is the minimum L value across all active rows in all previous blocks;
- $\text{flag} \in \{\text{true}, \text{false}\}$ indicates whether the current row is in a contiguous block.

Proof of Proposition 1.5. The boundary pair $(L_{\min}, L_{\min,\text{prev}})$ ranges over $\{(a, b) : 0 \leq b \leq a \leq n\} \cup \{(\bot, \bot)\}$, giving $\binom{n+2}{2}$ values (verified by `native_decide` for $n \leq 8$). The full state additionally tracks the minimum right-endpoint profile across the current contiguous block; the total reachable state count grows as the sequence 2, 6, 14, 25, 41, 63, 92, 129. Each transition corresponds to choosing a row state (empty or an interval $[L, R]$), checking compatibility with the constraint tuple, and updating the state. The computation of $|\text{CC}([m] \times [n])|$ reduces to m steps of matrix-vector multiplication with the transition matrix. □

This yields the first 50 terms of the square-grid sequence; Table 2 lists the first 10.

7 Comparison with Fuss–Catalan Numbers

The generalized Fuss–Catalan numbers with parameters $(p, r) = (6, 2)$ are

$$\text{FC}(m) \;=\; \frac{2}{6m+2}\binom{6m+2}{m},$$

m	$\lvert\mathrm{CC}([m]^2)\rvert$	$\lvert\mathrm{Int}([m]^2)\rvert$	a_m/a_{m-1}	$a_m/\mathrm{FC}(m)$
1	2	1	—	1.000
2	13	9	6.50	1.000
3	114	36	8.77	1.000
4	1,146	100	10.05	0.997
5	12,578	225	10.98	0.999
6	146,581	441	11.65	1.009
7	1,784,114	784	12.17	1.024
8	22,443,232	1,296	12.58	1.045
9	289,721,772	2,025	12.91	1.071
10	3,818,789,778	3,025	13.18	1.101

Table 2: First 10 terms of $|\mathrm{CC}([m]^2)|$ with interval counts $|\mathrm{Int}([m]^2)| = (m(m+1)/2)^2$, consecutive ratios, and comparison with $\mathrm{FC}(m) = \frac{2}{6m+2}\binom{6m+2}{m}$.

giving the sequence $2, 13, 114, 1150, 12586, 145299, \ldots$ (OEIS A212071 [9]). The first three terms coincide exactly with a_m:

$$a_1 = \mathrm{FC}(1) = 2, \quad a_2 = \mathrm{FC}(2) = 13, \quad a_3 = \mathrm{FC}(3) = 114.$$

The sequences diverge at $m = 4$: $a_4 = 1146$ while $\mathrm{FC}(4) = 1150$.

With 50 terms, the asymptotic behavior is clear: the ratio $a_m/\mathrm{FC}(m)$ diverges *exponentially*. At $m = 30$, $a_m/\mathrm{FC}(m) \approx 2.72$; at $m = 50$, $a_m/\mathrm{FC}(m) \approx 8.55$. The log of this ratio grows linearly in m with slope approximately 0.053, indicating that a_m has a strictly larger exponential growth rate than $\mathrm{FC}(m)$.

The Fuss–Catalan numbers have growth constant $6^6/5^5 = 46656/3125 \approx 14.9299$. Richardson extrapolation of the consecutive ratios of a_m yields estimates converging to the exact growth constant $\rho = 16$, which exceeds $6^6/5^5$ by approximately 7%.

Remark 7.1. The coincidence $a_m = \mathrm{FC}(m)$ for $m \leq 3$ is not accidental. For small m, the gapped sector is empty ($m \leq 2$) or negligible (5 subsets out of 114 at $m = 3$). The contiguous sector, which dominates at small m, appears to be counted by a Catalan-adjacent model. The divergence at $m = 4$ is the point where the multi-block sector first contributes significantly.

8 The Partition Function Interpretation

The block decomposition admits a natural statistical-mechanical interpretation.

Definition 8.1. The *block-weighted partition function* is

$$Z_m(\alpha) := \sum_{\substack{S \subseteq [m]^2 \\ S \text{ convex}}} e^{-\alpha \cdot k(S)},$$

where $k(S)$ is the number of contiguous blocks of S.

At $\alpha = 0$, $Z_m(0) = |\mathrm{CC}([m]^2)|$ (unweighted counting). As $\alpha \to \infty$, only the 0-block (empty set) and 1-block (contiguous) sectors survive.

Definition 8.2. The *free energy per row* is $f(\alpha) := \lim_{m\to\infty} \frac{1}{m} \log Z_m(\alpha)$, when the limit exists.

The supermultiplicativity theorem gives $f(0) = \log \rho \approx 2.77$. The contiguous sector has its own growth rate $\rho_{\text{contig}} \approx 12.8$, giving $f(\infty) \approx 2.55$. The *gap entropy* $f(0) - f(\infty) \approx 0.22$ measures the excess entropy contributed by multi-block configurations.

The block count $k(S)$ is not an arbitrary parameter: by Proposition 4.2, it equals the number of comparability-connected components of the active rows, a structural invariant forced by the classification theorem.

The Noetherian ratio relates to the partition function as

$$\gamma([m]^2) \;=\; \frac{Z_m(0)}{|\mathrm{Int}([m]^2)|} \;=\; \frac{Z_m(0)}{(m(m+1)/2)^2}.$$

9 Formal Verification

Theorems 1.1–1.3, Corollary 1.4, and the upper bound $\rho \le 16$ are fully verified using the Lean 4 proof assistant with Mathlib [8]. The lower bound $\rho \ge 16$ (Theorem 10.2) and the dimension law (Theorem 10.4) are fully verified. Exact counts for $[m]^2$ with $m \le 4$ are kernel-verified; values for $m \ge 5$ are computed by the transfer-matrix algorithm. Source code is available at [11].

10 Conjectures

Remark 10.1. The generating function $A(z) = \sum_{m\ge1} |\mathrm{CC}([m]^2)|\, z^m$ is likely *not* D-finite. A systematic search found no recurrence of order ≤ 5 with polynomial coefficients of degree ≤ 4 satisfied by the first 50 terms. This is consistent with a natural boundary in the generating function and explains the difficulty of the exact enumeration problem: no simple closed-form recurrence exists.

Theorem 10.2. *The growth constant is $\rho = 16 = 2^4$ exactly. Both bounds are fully verified in Lean 4 [11] with zero* `sorry` *and zero custom axioms.*

Proof. Upper bound. Every convex subset $S \subseteq [m]^2$ is determined by a pair (D, U) of antitone functions $D, U\colon [m] \to [m{+}1]$ representing its downward and upward boundary (the "band set" encoding). The map $S \mapsto (D, U)$ is injective. Each antitone function $[m] \to [m{+}1]$ is counted by $\binom{2m}{m}$ via stars-and-bars. Hence $|\mathrm{CC}([m]^2)| \le \binom{2m}{m}^2 \le 16^m$.

Lower bound. A pair (D, U) with $D(i) > U(i)$ for all i (a "non-crossing pair") maps injectively to a convex subset. The number of crossing pairs is bounded by $\binom{2m+1}{m} \cdot \binom{2m-1}{m}$ via a Lindström-type domain-splitting injection: given a crossing pair with first crossing at index i_0, concatenate the U-prefix with the D-suffix (and vice versa) to produce antitone functions of domain sizes $m+1$ and $m-1$. Injectivity follows because distinct crossing indices lead to contradictions with antitonicity. The algebraic identity $\binom{2m}{m}^2 - \binom{2m+1}{m}\binom{2m-1}{m} = \binom{2m}{m}^2/(2(m{+}1))$ yields $|\mathrm{CC}([m]^2)| \ge \binom{2m}{m}^2/(2(m{+}1)) \ge 16^m/\mathrm{poly}(m)$.

Squeeze. Supermultiplicativity (Theorem 1.3) and Fekete's lemma give $\rho = \lim |\mathrm{CC}([m]^2)|^{1/m}$. The upper bound gives $\rho \le 16$. The lower bound gives $|\mathrm{CC}([m]^2)|^{1/m} \ge 16 \cdot (2(m{+}1))^{-1/m} \to 16$. Hence $\rho = 16$. □

Conjecture 10.3. *The subexponential factor satisfies*

$$|\mathrm{CC}([m]^2)| \;\sim\; C \cdot 16^m \cdot m^{-\alpha}$$

for constants $C > 0$ and $\alpha > 0$.

Theorem 10.4. *For d-dimensional grids $[m]^d$ with $d \ge 2$, the sequence $|\mathrm{CC}([m]^d)|$ is supermultiplicative and $\log|\mathrm{CC}([m]^d)| = \Theta(m^{d-1})$. For $d = 2$, the growth constant $\rho_2 := \lim_{m\to\infty} |\mathrm{CC}([m]^2)|^{1/m} = 16$ (Theorem 10.2). For $d \ge 3$, the $1/m$ normalization diverges; the correct normalization is $c_d := \lim_{m\to\infty} \log|\mathrm{CC}([m]^d)|/m^{d-1}$, which exists by Fekete's lemma applied to the superadditive sequence $\log|\mathrm{CC}([m]^d)|$.*

Proof. Supermultiplicativity for all $d \geq 2$ is proved by the same concatenation argument as Theorem 1.3. *Upper bound:* $|\mathrm{CC}([m]^d)| \leq |\mathrm{downsets}([m]^d)|^2$ by the same ideal/filter injection. *Lower bound:* Partition $[km]^d$ into k^d blocks via the embedding $f \mapsto \alpha \cdot m + f$ for $\alpha \in [k]^d$. Points from incomparable blocks (i.e., $\alpha \| \alpha'$ in the product order on $[k]^d$) are incomparable in $[km]^d$. The grid $[k]^d$ has antichains of size $\Theta(k^{d-1})$; placing independent convex subsets in antichain-positioned blocks yields $|\mathrm{CC}([km]^d)| \geq |\mathrm{CC}([m]^d)|^{\Theta(k^{d-1})}$. Combined: $\log |\mathrm{CC}([m]^d)| = \Theta(m^{d-1})$. Formally verified; see [11]. □

Conjecture 10.5. *There exists a bijective encoding of order-convex subsets of $[m]^2$ by pairs of monotone boundary paths with anti-monotone gap constraints, giving a combinatorial interpretation of the counting sequence.*

11 Discussion

The central finding of this paper is strong numerical evidence that order-convex subsets of grid posets form a new exponential combinatorial class, with a growth constant distinct from all known Catalan/Raney families. The structural explanation is the two-sector decomposition: contiguous blocks behave in a Catalan-like manner, while anti-monotone gaps introduce an independent combinatorial layer that changes the leading growth rate.

The coincidence $a_m = \mathrm{FC}(m)$ for $m \leq 3$ is explained by the fact that the gapped sector is empty or negligible for small m. The divergence at $m = 4$ marks the onset of the multi-block sector, which grows to comprise 23% of the total by $m = 10$ and drives the exponential separation from the Fuss–Catalan curve.

The supermultiplicativity theorem, whose proof uses the same anti-monotone embedding mechanism that generates the new class, converts the growth constant from numerical evidence to a theorem-level statement.

Several directions remain:

1. *Determine the subexponential factor*: prove or disprove $|\mathrm{CC}([m]^2)| \sim C \cdot 16^m \cdot m^{-\alpha}$ and identify the exponent α.
2. *Derive a recurrence or functional equation* from the transfer-matrix specification.
3. *Extend to* $[m] \times [n]$ and compute the bivariate generating function, which may reveal the true combinatorial mechanism more clearly than the square case.
4. *Determine exact constants c_d for $d \geq 3$*: the dimension law $\log |\mathrm{CC}([m]^d)| = \Theta(m^{d-1})$ is proved (Theorem 10.4), but the exact constants $c_d = \lim \log |\mathrm{CC}([m]^d)|/m^{d-1}$ for $d \geq 3$ are open.

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The formal verification builds on the Mathlib library [8].

References

[1] G. Birkhoff, *Lattice Theory*, 3rd ed., AMS Colloquium Publications, vol. 25, 1967.

[2] B. Barnette, W. Nichols, and T. Richmond, The number of convex sets in a product of totally ordered sets, *Rocky Mountain J. Math.* **49** (2019), no. 2, 369–385.

[3] L. Bombelli, J. Lee, D. Meyer, and R.D. Sorkin, Spacetime as a causal set, *Phys. Rev. Lett.* **59** (1987), no. 5, 521–524.

[4] T. DiFiore, Causal-algebraic geometry: A Grothendieck-type framework for partial orders, preprint, 2026.

[5] R.P. Dilworth, A decomposition theorem for partially ordered sets, *Ann. of Math. (2)* **51** (1950), 161–166.

[6] M. Fekete, Über die Verteilung der Wurzeln bei gewissen algebraischen Gleichungen mit ganzzahligen Koeffizienten, *Math. Z.* **17** (1923), 228–249.

[7] G. Grätzer and R.W. Quackenbush, Positive universal classes in locally finite varieties, *Algebra Universalis* **64** (2010), 1–13.

[8] The Mathlib Community, *Mathlib: The Lean Mathematical Library*, https://github.com/leanprover-community/mathlib4, 2024.

[9] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2024.

[10] M.V. Semenova and F. Wehrung, Sublattices of lattices of order-convex sets, II: Posets of finite length, *Internat. J. Algebra Comput.* **13** (2003), no. 5, 543–564.

[11] T. DiFiore, *Causal-Algebraic Geometry in Lean 4*, https://github.com/tomdif/causal-algebraic-geometry-lean, 2026.